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ENG 1102 OT2T4

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OT2T4_Phase1_Report

1. Problem Identification and Context

1.1 Introduction

Recently, Toronto has been facing a significant challenge, with the snow accumulation reaching 65cm. This level of snow has transcended the typical winter inconvenience, escalating the situation into a huge disturbance of daily life. The frequent blizzards have created a cumulative effect, as the snow continues to pile up, and ongoing manual snow removal is required, leading to a certain blockage in the residential areas. For the average citizen or house owner, without doubt, it is consuming a huge amount of time and energy. However, for the elderly and the disabled, it heavily hinders their basic mobility needs.

1.2 Driveway:

One of the biggest issues is the impact on the residential driveway, which is the pathway for people to enter or exit their house. And it is now presenting itself as a unique physical hazard. Unlike the sidewalks, which are usually flat surfaces, driveways are often built on an incline. When a vehicle runs over it, it compresses the snow into a thin, icy layer that covers the driveway, making it nearly impossible to navigate safely. Especially in the evening, when the lights are dim, people can barely notice the ice on the driveway. For residents attempting to clean up the area, carrying heavy objects on the icy surface can easily lead to slips and falls. What's worse is that for the elderly and disabled, it wouldn't be just a minor injury; it is something that can lead to permanent harm or even fatal consequences.

1.3 Evidence:

On January 7, 2026, an 80-year-old Toronto woman was struck by a vehicle after slipping and falling on ice. According to preliminary information, the truck was backing up the driveway when the woman was walking past and fell down. This tragic event, which begins with a slip,

showing a pedestrian losing their footing in a dangerous zone, leads to becoming a victim of the situation.

Another event that happened in Edmonton highlights the story of Kathleen Joy, who broke her ankle in a "split second" while brushing snow off her car. Despite her attempts to grab the vehicle to maintain her balance, the slippery ice caused her to suddenly fall heavily, becoming trapped inside and shouting for help for several minutes.

1.4 Why is it important:

The issue is a fundamental breakdown of public safety and mobility. The current method of relying on the individual property owners to manage record-breaking snowfall is insufficient, as we saw from above, how, without proper management of the driveway, it could lead to potential fatal mistakes. A city's infrastructure is a failure if a large portion of its population cannot safely leave their homes to access food, medication, or emergency care.

2. User Understanding and Empathy

2.1 Who are the Users

The problem of snow on driveways is a very different situation based on several factors on several different users. Narrowing it down to our target users, we define them as users who are seniors of 65 years or older, located in Ontario, who need to deal with snow and ice on their driveway. The reason for this specific age is that someone of a younger age may see this problem as a minor inconvenience of their day, but people of older age see this as a serious problem with serious issues. The reason Ontario is the location is because of the large and heavy snowfall that has been occurring in the past few years during the winter seasons.

2.2 Methods Used to Understand Users

User Interviews:

We conducted several interviews with neighbours who were leaning towards older age, asking in-depth questions about their experiences managing property maintenance in winter

Online Surveys:

We created an online survey to be shared with several homeowners across different regions in Ontario, getting over 50 responses

Observations:

We observed this mass problem in everyone's life from looking out our windows, on a drive around the city, and everywhere we go.

Secondary Sources:

Read over the statistics and reports that were conducted by a subgroup of the Government of Canada. These helped us understand that this is an actual problem and not just an inconsistency.

2.3 Key User Insights

From the analysis of our interview responses and survey results. Three key use insights were identified. Each of these insights reflects a clear tension between seniors who need to safely maintain their property during winter and the limitations of current snow removal methods. These findings help explain that snow problems on driveways are not simply an inconvenience, but a meaningful and ongoing problem.

Insight 1: Snow removal causes physical strain, while current methods only offer short-term relief.

In the interviews, seniors explained that clearing snow has become more physically demanding with age. Tasks that once felt routine now result in back pain, fatigue, and soreness afterwards. Many surveyors responded that they spend 20 or more minutes shovelling, often feeling tired before finishing the task. Although tools such as shovels, salt, and snow blowers are commonly used, they only provide temporary improvement, as snow and ice frequently return, then they would have to do the same effort and take the same risks again. This creates tension between the seniors' desire to remain independent and their decreasing physical ability to repeatedly manage winter maintenance.

Insight 2: The fear of slipping creates ongoing stress and safety concerns.

A large portion of survey participants report that they have experienced, witnessed, or nearly experienced a slip or fall due to snow and ice. During knocking, several seniors expressed concern about falling, particularly in the early morning or evening when visibility is limited. For older adults, a fall can lead to serious injury. In fact, falls are the leading cause of injury-related hospitalisations among elders in Canada. This highlights a conflict between the need to safely access one's home and the constant anxiety caused by icy driveway conditions

Insight 3: Existing solutions are viewed as repetitive and unreliable

Many users describe their frustration with the constant cycle of clearing snow only for it to return within days or even hours. While salting and shovelling may temporarily improve conditions, freezing temperatures often cause ice to reform quickly. Some seniors also mentioned that city services or assistance are not always timely or even consistent. This demonstrates an unmet need for a reliable approach that reduces repetition and effort. The reliance on temporary solutions contributes not only to physical exhaustion but also makes seniors tired and frustrated throughout the winter.

3. Problem Statement and Criteria for Success

3.1 Problem Statement

Seniors aged 65 and older in Ontario struggle to keep their driveways and outdoor paths clear during the winter. Snow and ice make shovelling hard and sometimes dangerous, especially in cold or slippery conditions. Current methods of removing snow, like manual shovelling or relying on city services, are usually slow, difficult or unreliable, leaving many seniors at risk and making it harder for them to maintain their property.

3.2 Criteria for Success

Scope:

Our project will focus on single detached or semi-detached home solutions that enable the clearing of snow and ice from driveways, walkways, and porches with minimal or no physical effort required from the user. Our solution must be suitable for Canadian winter conditions, including heavy snowfall, ice accumulation, and extreme cold, and should be capable of operating during or immediately after snowfall to ensure timely and safe access. We will not

be addressing the larger-scale removal of snow from parking lots, parks, or other examples of large-scale public lots. Our ideal solutions won't require extensive home renovations or continuous labour from users or contractors.

Requirements:

Affordability: Traditional heated driveways require the removal and replacement of existing driveways, resulting in high upfront costs. In Ontario, these systems typically cost between \$3,000 and \$8,700; a key requirement of our solution is to remain below this price range.

Usability: Our solution must be easy to use for elderly individuals, thus it should require minimal physical effort to operate. Additionally, controls and instructions should be simple, and the solution should ideally have a simple setup.

Scalability: The solution should be able to accommodate a wide range of driveways and residential entrances. It should be easily adjustable or modular so that it can be adapted to different home layouts without requiring major redesigns or customization.

Constraints: Project constraints include limited access to materials beyond the EECS lab kit. Additionally, our solution ideation and prototyping must be completed within a single academic term.

3. References and Appendices

4.1 Appendix A: Interview Questions

To get a better understanding of our users, we conducted in-person interviews with various neighbours across the North York region. These questions were designed to understand the homeowners of older ages' perspective of property maintenance, especially in the winter

season. We sought answers on how the struggle of maintaining snow disrupted them physically, mentally, and what kind of solutions they have turned to in the past. The goal was to understand the overall impact of our problem, specifically for these households and to determine if this is an underlying problem or just a small inconvenience.

Interview Questions Asked
How do you currently manage snow removal on your property?
Have you ever felt unsafe or physically strained while clearing snow in the winter months?
Can you describe a specific challenge you face while clearing snow?
How much time does it usually take for you to clear the snow?
What factors are most important to you when selecting a snow removal method?

4.2 Appendix B: Interview Summaries

The in-person interviews revealed consistently similar responses from all the homeowners as we spoke with them. Physically, a lot of them said that as their age starts increasing, tasks that they could do easily many years ago become extremely difficult. Several interviewees showed concern for their safety, describing that in the morning, when it's dark, the surroundings are extremely hard to see, and it is very possible to get seriously injured. Many have tried different solutions, such as city services and snow blowers, but that doesn't stop the constant snow, and these methods were very unreliable. Overall, a lot of the homeowners expressed a significant interest in different solutions that they have not tried yet.

4.3 Appendix C: Google Form Survey Results

To gather more responses from homeowners from different neighbourhoods, not only from North York, but we also launched a Google form that was filled out by more than 50 people. Most reported that they spent 15-30 minutes shovelling snow and getting tired halfway before they were finished. Then we were interested in what the majority of people have experienced: a slip or a fall on their driveway while shovelling. Overall, the responses for the other questions were very similar to the ones retrieved from the in-person interviews.

4.4 Appendix D: Empathy Map

Based on the feedback gathered through interviews and surveys, we created an empathy map that displayed how these older homeowners really feel about our problem in a very organized manner.

4.4.1 Table 1 Empathy Map(Says, Does, Thinks)

Says	Does	Thinks
“The driveway gets really slippery after it snows.” “I worry about falling, especially when it is dark.” “The snow keeps coming back even after I shovel.”	Wakes up early to shovel snow before family members leave for the day Shovels snow, ignoring temperatures and pain Clears only the most necessary path due to fatigue	Worries about slipping and getting seriously injured Thinks if the shovelling process is snow family will be late Thinks removing snow is becoming harder with age

4.4.2 Table 2 Empathy Map(Feel, Pain, Gain)

Feel	Pain	Gain
Stressed out about making	Physical pain after cleaning	Reduced effort allows energy

everyday tasks harder Discouraged by the effort needed for snow removal	snow in cold weather Lack of alternatives	to be saved for the rest of the day Less anxious about stepping outside
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4.5 References

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4. Team Contribution Table

Team Member Name	Type of Task	Specific Task	Approx Time/ Effort
#1 Herman	Research, Teamwork	Appendices, 2.1, 2.2, references	2.5 hours/ High
#2 Yui Him	Research, organize	1.1-1.4, references	2.5 hours/ High
#3 Berke	Team supporter, Research	3.1, 3.2	2 hours/ High
#4 Tawil	Idea, Research	2.3, references	2 hours/High

#5 Matteo	Overall research		1 hour
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